

collection of programs $\rightarrow$ software package.

A program will have variables of some types.

`main(){} → does compile -`

`gcc → has flags`

`--save-temps → save intermediate files.`

Where are the imports in C (include) located?

`bin → binary folder`

$\rightarrow$ has compiled blocks (?)

c) what are the analysis tools available to analyze ELF files?

intermediate compile files $\rightarrow$

emptyProg.i

emptyProg.s $\rightarrow$ assembly $\rightarrow$ even though it's an empty file, it has lots of content, as it needs to do stuff like starting a process.

emptyProg.o

d) what is a relocatable binary?

e) what are the properties associated with a file?

(readable, executable, ...)

$$\text{Software} = \bigcup_{i=0}^n \text{program}_i$$

program $\rightarrow$ / logic + control + D/S /

(in a source language)

$\hookrightarrow^*$
executable

a) Can we compile a python script?

"build" tab in editors

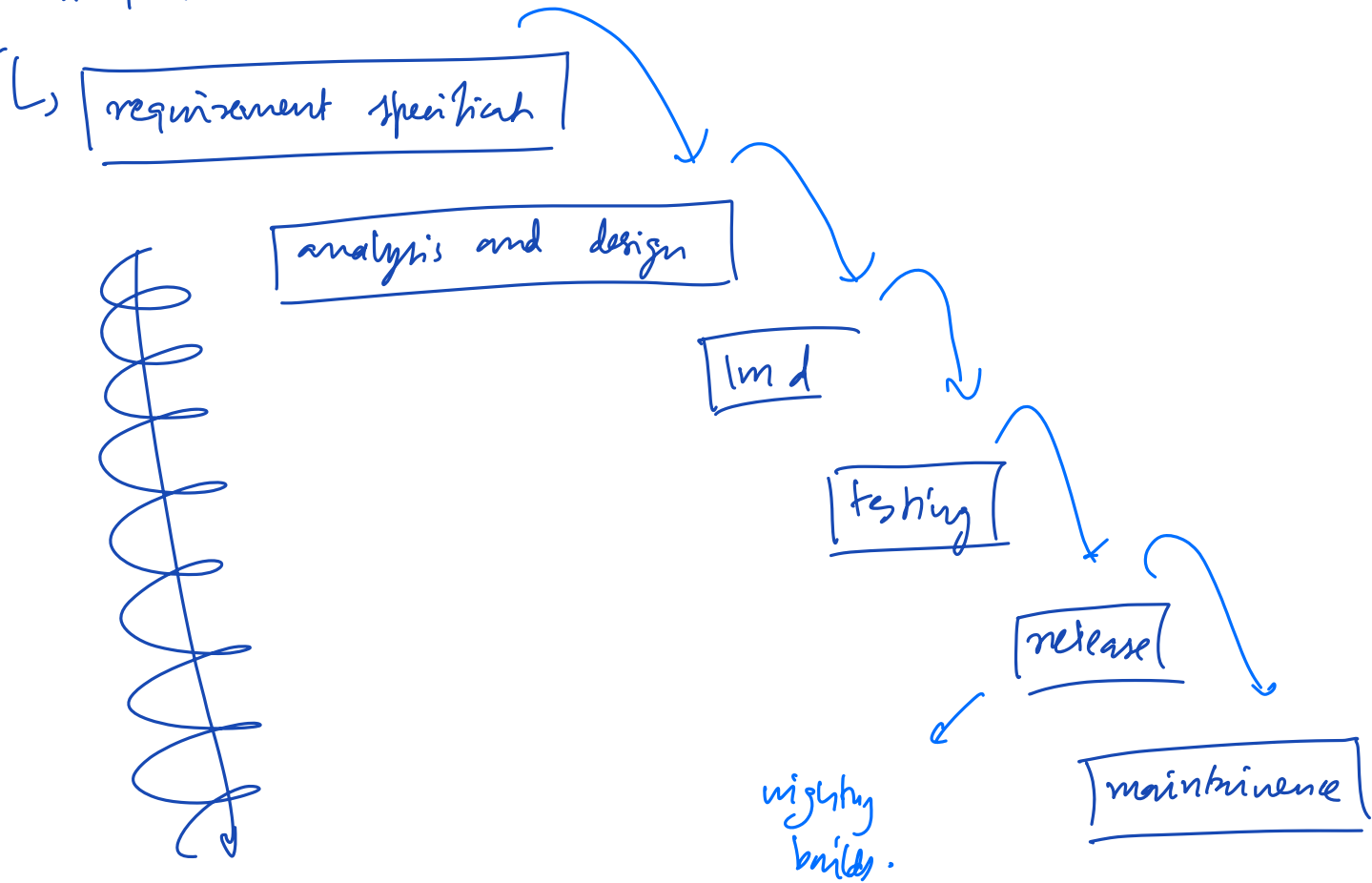
build script is formed as you write the program, and is executed

"Build Systems"

a) Identify the most popular build systems

→ Waterfall model

boxes -



other.

→ spiral version

→ C/C++

program

$$X = \{x_1, x_2, \dots, x_n\}$$

$\langle i_1, i_2, \dots, i_n \rangle \rightarrow$ initial variables

program
counter

$\langle i_1', i_2', \dots, i_n' \rangle$

state of a program $\rightarrow$ binding of all variables

Theorem providers:

Model checkers: SPIN, Uppaal

Static Analysers: Astrée